

BUSINESS PROBLEM

The telecommunications company is experiencing customer churn, resulting in the loss of recurring revenue and the reduction of customer lifetime value. The objective of this analysis is to identify the customer characteristics, services, contract types, and billing factors that is associated with churn, so the company can understand why customers leave and develop a good retention strategy.

Question 1

Why are customers leaving the company?

Question 2

Which factors contribute most to customer churn?

DATASET DESCRIPTION

The WA_Fn-UseC_-Telco-Customer-Churn dataset contains 7043 rows and 21 columns, including: customerID, gender, SeniorCitizen, Partner, Dependents, tenure, PhoneService, MultipleLines, InternetServices, OnlineSecurity, OnlineBackup, DeviceProtection, TechSupport, StreamingTV, StreamingMovies, Contract, PaperlessBilling, PaymentMethod, MonthlyCharges, TotalCharges, and Churn.

Dataset Evaluation

There are 1869 churned customers out of the 7043 customers, generating a churn rate of 26.54%.

The following hypothesis were evaluated to find the possible cause to why customers left the company:

- Hypothesis One: Charges and Customer Tenure
If churned customers have higher Monthly Charges, this may suggest that the cost of services or the perceived value could be contributing to customers leaving. If churned customers have substantially lower tenure, this would suggest that customers are more likely to leave during the early stages of their partnering with the company.
 - Observation:
 - 1. Churned customers have shorter tenure.**
 - Customers who leave tend to be newer customers.
 - This suggests the company has difficulty retaining customers during the early stages of the customer relationship.
 - 2. Churned customers pay more per month.**
 - The average monthly charge of churned customers is **\$74.44**, compared with **\$61.27** among retained customers.

- That's a difference of **\$13.17 per month**, or about **21.5% higher** for churned customers.
- Deduction:

Customers may be leaving relatively early in their partnering with the company while paying relatively high monthly charges. Newer customers may be more sensitive to the price they are paying per month relative to the value they perceive from the service.
- Hypothesis Two: Why are churned customers paying \$74.44 while retained customers are paying \$61.27?

The analysis of monthly charges by internet services showed that there was an inconsistency in the monthly charge. Though the churned customers had a higher monthly charge, this difference wasn't consistent across the individual internet-service categories.

 - Observation: Retained customers had higher average monthly charges than churned customers across all three internet-service categories. Fiber-optic customers who remained with the company paid an average of **\$93.93** per month, compared with **\$88.13** among churned customers. Similarly, retained DSL customers paid **\$60.21** compared with **\$49.08** among churned customers.
 - Deduction: Higher monthly charges alone do not adequately explain customer churn. The higher overall monthly charge observed in the churned customers appears to be influenced by the composition of the churned customer group, as approximately **69% of churned customers use fiber-optic service**, compared with approximately **35% of retained customers**.
- Hypothesis Three: Fiber Optic Customers: Churn Rate by Contract Type

The previous analysis identified fiber-optic customers as a large proportion of the churned customer base. To further investigate this, I examined the churn rate of fiber-optic customers across different contract types.

 - Observation: The previous analysis identified fiber-optic customers as an excessively large proportion of the churned customer base. To investigate this further, I examined the churn rate of fiber-optic customers across different contract types.
 - Deduction: Among the customers using fiber-optic internet service, churn is strongly associated with the contract type. Month-to-month fiber-optic customers have a churn rate of 54.61%, compared with 19.29% for one-year contracts and 7.23% for two-year contracts. This indicates that the

lack of a long-term contractual commitment may be an important characteristic of customers who leave the company.

- Hypothesis Four: Is lack of support contributing to churn among high-risk customers?

Customers who lack Technical Support and Online Security may be more likely to churn, particularly among fiber-optic customers who run on month-to-month contracts.

- Observation: The analysis of this high-risk segment revealed a clear relationship between access to Technical Support/Online Security and customer churn:

Technical Support	Online Security	Churn Rate
No	No	60.70%
Yes	No	42.40%
No	Yes	39.71%
Yes	Yes	28.05%

Customers who had **neither Technical Support nor Online Security** had the highest churn rate at **60.70%**. In contrast, customers who had **both services** (technical support and online security) had the lowest churn rate at **28.05%**.

This represents a **32.65 percentage-point difference** in churn between the two groups.

- Deduction: These results suggest that the absence of Technical Support and Online Security is strongly associated with higher customer churn among fiber-optic, month-to-month customers. Customers who have access to both services demonstrate substantially lower churn than customers who have neither service.

- Hypothesis Five: Payment Method Among the Highest-Risk Group
The previous hypotheses identified fiber-optic customers on month-to-month contracts without Technical Support or Online Security as the highest-risk segment, with a churn rate of 60.70%.

This analysis is to investigate whether payment method is also associated with churn in this high-risk group.

- Observation: The analysis produced the following results:

Payment Method	Total Customers	Churned Customers	Churn Rate
Electronic check	989	651	65.82%
Mailed check	137	76	55.47%
Bank transfer (automatic)	212	110	51.89%
Credit card (automatic)	186	88	47.31%

The highest churn rate was observed among customers using **electronic check**, at **65.82%**.

This was followed by:

- Mailed check = **55.47%**
- Bank transfer = **51.89%**
- Credit card = **47.31%**

There is therefore a **18.51 percentage-point difference** between electronic check and credit card customers within this high-risk segment.

- Deduction: Payment method shows to be strongly associated with churn within the highest-risk customer segment. Customers using electronic checks had the highest churn rate at 65.82%, compared with 47.31% among customers using automatic credit-card payments.

This suggests that customers using electronic checks may represent an especially vulnerable group and could potentially experience a different billing or payment experience compared with customers using other payment methods.

OVERALL FINDINGS

Based on all five hypothesis that have been carried out, the analysis indicates that customer churn is concentrated among relatively **newer customers** and is particularly common among fiber-optic customers. Although churned customers have a higher overall average monthly charge, further analysis showed that retained customers

actually pay more than churned customers within each internet-service category. This suggests that higher monthly charges alone do not explain customer churn.

The strongest pattern emerged among **fiber-optic customers on month-to-month contracts**, who had a churn rate of **54.61%**. Within this group, customers without Technical Support and Online Security had an even higher churn rate of **60.70%**.

Among this highest-risk group, customers using **electronic checks had the highest churn rate at 65.82%**.

These findings suggest that customer churn is associated with a combination of factors rather than a single cause. In particular, shorter tenure, month-to-month contracts, fiber-optic service, lack of additional support/security services, and electronic-check payment methods are characteristics that are strongly associated with higher churn in this dataset.

RECOMMENDATIONS

1. Prioritize the retention of customers who use the fiber-optic internet service, and are month-to-month customers.
2. Onboarding and early-stage customer retention should be strengthened.
3. Access to technical support and online security should be increased.
4. Investigation on why electronic-check customers have substantially higher churn should be carried out.

CONCLUSION

The analysis indicates that the customer churn is associated with a combination of factors rather than a single issue. Firstly, churn is particularly concentrated among shorter-tenured customers, fiber-optic users, and customers on month-to-month contracts, with churn increasing more among those without Technical Support or Online Security and those using electronic checks. Although churned customers have higher overall monthly charges, the analysis shows that higher charges alone do not explain churn, as retained customers pay more within each internet-service category. Overall, the findings suggest that the company should focus on early customer retention, stronger support and security services, longer-term contract adoption, and targeted intervention for high-risk customer segments rather than relying on broad price reductions.